Bellabeat Data Analysis

INTRODUCTION

Urška Sršen and Sando Mur founded Bellabeat, a high-tech company that manufactures health-focused smart products. Sršen used her background as an artist to develop beautifully designed technology that informs and inspires women around the world. Collecting data on activity, sleep, stress, and reproductive health has allowed Bellabeat to empower women with knowledge about their own health and habits. Since it was founded in 2013, Bellabeat has grown rapidly and quickly positioned itself as a tech-driven wellness company for women.
Source: Google Data Analytics Professional Certificate, Bellabeat case-study packet.

BUSINESS TASK

Bellabeat would like me to take one of its products and analyze smart device data in order to gain insight into how consumers are using their smart devices. These insights will help recommend a marketing strategy for the company. The product in question will be Bellabeat's Leaf which tracks activity, sleep, & stress.

STAKEHOLDERS

- Urška Sršen: Bellabeat's cofounder and Chief Creative Officer (Requested analysis).
- Sando Mur: Mathematician and Bellabeat's cofounder; key member of the Bellabeat executive team (Will consider recommendations).
- Bellabeat marketing analytics team: A team of data analysts responsible for collecting, analyzing, and reporting data that helps guide Bellabeat's marketing strategy (Supports strategy with evidence).

ASK:
1. What are some trends in smart device usage?
2. How could these trends apply to Bellabeat customers?
3. How could these trends help influence Bellabeat marketing strategy?

Prepare:

The data that will be used to explore smart device users' daily habits is public data from https://www.kaggle.com/datasets/arashnic/fitbit FitBit Fitness Tracker Data which is under the CC0: Public Domain made available by Mobius. Two datasets were downloaded from the FitBit Fitness Tracker Data. Both datasets contain records dated April 12 through May 12, 2016 which are considered historical as habits and devices may not reflect current behavior. Within the dataset DailyActivity_merged.csv 33 distinct participants tracked their Daily Activity. Originally there were 940 rows of data which remained unchanged after cleaning. Within the dataset SleepDay_merged.csv 24 participants tracked their Daily Sleep over the same period of time. Originally there were 413 rows of data, after cleaning there were 410 rows of data after removing 3 duplicates. Also, Calories are included within the DailyActivity_merged.csv dataset. Because Bellabeat is a company that primarily focuses on women, the data itself does not directly mention if the participants are indeed women, please keep this in mind for the final results of the data. Also our case study describes using 30 users for this activity, the selected activity file does contain 33 distinct IDs.

Process: Before processing and manipulating the data, the datasets were uploaded to Google Sheets where the following findings were revealed:

- Removed 3 duplicate rows from the sleep dataset by comparing all five original columns, retaining one copy of each record. The dataset decreased from 413 to 410 rows. The remaining data checks required no change.
- Missing participant-days remained missing, retained zero-step records, and excluded the four zero-calorie records only for the sensitivity calculation.
- Checked minimum and maximum dates.
- Checked numeric storage in the selected date and measurement fields.
- Found no blank cells in either original data range.
- Found no negative measurements in the ranges checked.
- Confirmed no repeated participant–date combinations after cleaning.
- **DayOfWeek:** Created from ActivityDate using "TEXT" to group records by weekday.
- **MinutesSum1440:** Flagged whether VeryActiveMinutes, FairlyActiveMinutes, LightlyActiveMinutes, and SedentaryMinutes totaled 1,440. Used to check whether the weekday findings changed under that restriction.
- Found no sleep records where minutes asleep exceeded time in bed.

Analyze: A scatter plot chart was created to identify the relationship between Total Steps and Estimated Calories Burned. Pearson's correlation measured its linear strength and direction. Higher recorded steps were associated with higher estimated calories burned with variation. The correlation turned out to be 0.592 across 940 records compared with 0.587 across 936 records after excluding zero calorie entries. The records include repeated observations from 33 participants so the correlation summarizes recorded days rather than 940 independent people which means the correlation doesn't establish causation.



Figure 1. Daily steps and estimated calories burned across 940 recorded participant-days from 33 participant IDs, April 12–May 12, 2016. Each point represents one participant's recorded day. Zero-value records are included.

Here's a Histogram of the recorded sleep durations. After conducting analysis, recorded sleep duration ranged from 58 to 796 minutes. The average sleep duration came out to 419.17 minutes & the median came out to 432.5 minutes. The middle 50% of recorded sleep durations fell between 361 and 490 minutes. In total there were 410 participant days from 24 participant IDs. It is worth noting that recorded sleep duration alone doesn't establish sleep quality or explain unusually short or long records.

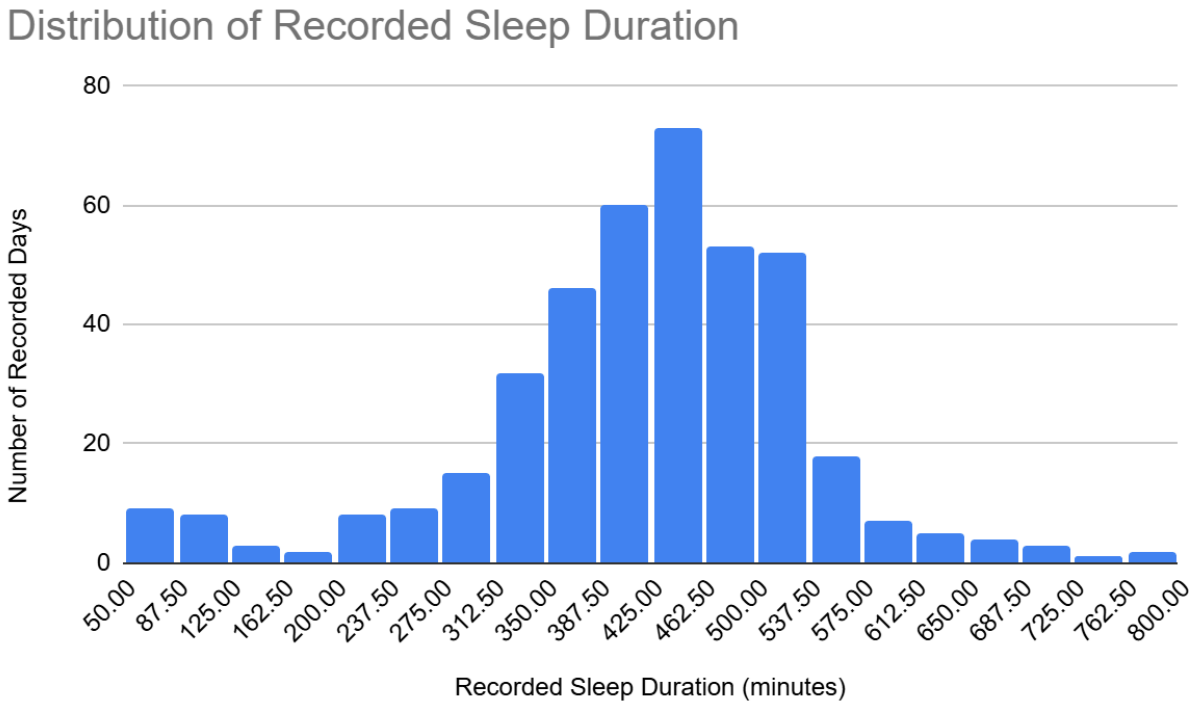


Figure 2. Distribution of recorded sleep duration across 410 participant-days from 24 participant IDs, April 12–May 12, 2016, after removing three duplicate records. Bar heights represent the number of records within each sleep-duration interval.

Daily Steps by Weekday: To document comparisons between Weekdays and Avg. steps per recorded day a bar chart was created to identify the relationship. Pivot tables were created to compare the averages. The following findings were out of the 940 records in the full dataset and 478 in the filtered subset, 1,440 minutes doesn't prove continuous wear. The biggest finding was that before filtering Saturday had the highest average, after filtering Saturday changed to the lowest. Because of this a recommendation targeting one specific weekday isn't stable across the two selections. It's worth noting the filter changes the mix of records and participants; it doesn't prove the retained records are more accurate. The filter also reduces representation from 33 to 28 participant IDs.

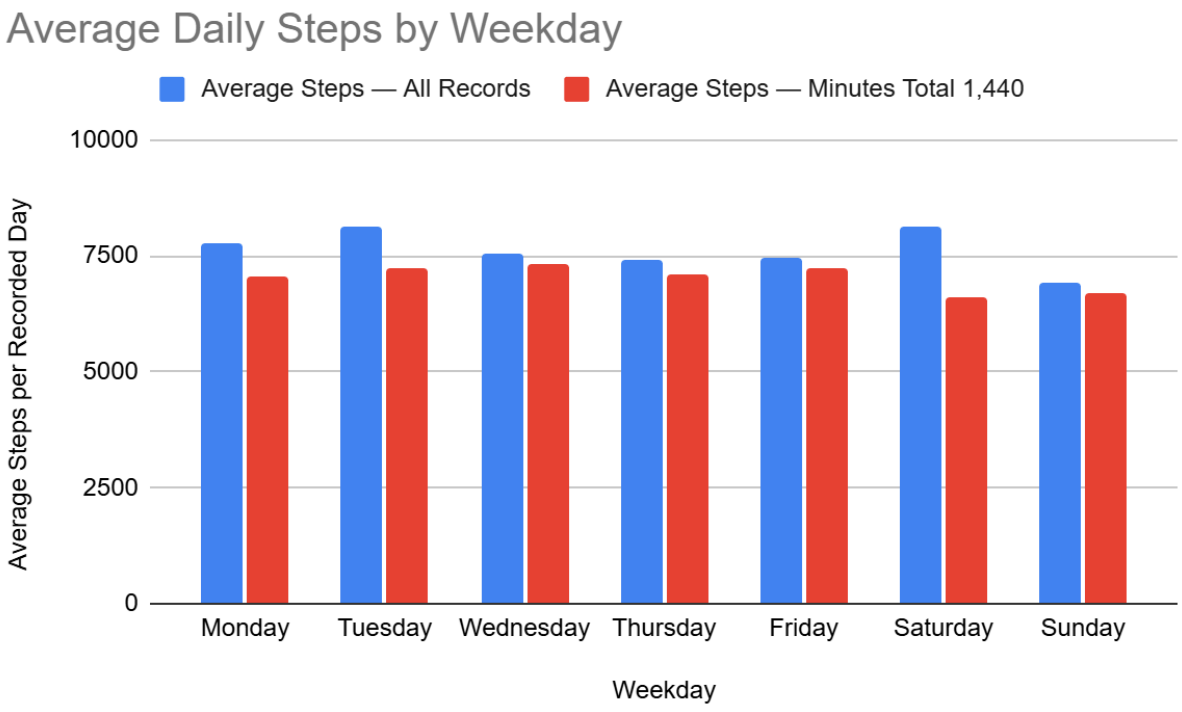


Figure 3. Average daily steps by weekday for all 940 activity records and the subset of 478 records whose four activity-minute fields total 1,440 minutes. The full sample contains 33 participant IDs; the subset contains 28. A 1,440-minute total does not establish continuous device wear.

Act:

1. Based on my findings, analysis showed a correlation between steps and estimated calories burned. My recommendation is to test messaging about understanding personal activity patterns such as helping users recognize changes in their daily movement. Also, compare the purchase conversion rate for the activity-pattern message with the existing message to assess whether it encourages more visitors to purchase Leaf.

   Purchase conversion rate = visitors who purchase Leaf ÷ total visitors to that message's landing page × 100.

2. Analyzing recorded sleep durations, we found that the median sleep time was 432.5 minutes, with the middle 50% between 361 and 490 minutes. Because there was an unequal number of records per participant, that can be seen as a limitation as participants with more recorded days contributed more to the summary. What could benefit Bellabeat is to test messaging about recognizing sleep-duration patterns over time and compare its purchase conversion rate with the existing message.

3. After analyzing the Average Total Steps throughout the week, we found the rankings of weekdays by average steps changed when the analysis was restricted to records whose minute fields totaled 1,440. We recommend that Bellabeat collect more inactivity data and user preferences. These include the following:

   - Inform reminder timing & compare personalized reminders with a fixed schedule.
   - Identify whether reminders increase subsequent movement over a defined period for both groups, and monitor dismissals or opt-outs.

Limitations:

After reviewing the two selected datasets we did find some limitations, here's the following:

- The volunteer sample was small (33 activity IDs & 24 sleep IDs) for the Fitbit sample with unknown relevance to Bellabeat's target customers.
- Upon examining the data it was realized that the data was historical, being from April–May 2016 which limits conclusions about long-term habits and current users.
- Participants with more recorded days contributed more towards the summary and absent days may differ from recorded days.
- Calories are estimates and a 1,440-minute total doesn't prove continuous wear. Uncertain coverage limits confidence in the activity comparisons used to inform recommendations.
- No direct evidence about Leaf's accuracy, stress, sleep quality, or marketing effectiveness.